

TABRIZ INTL, Iran, Islamic Republic of

WMO: 407060

Lat: 38.134N

Lon: 46.235E

Elev: 1359

StdP: 86.03

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-9.1	-17.1	1.0	-7.8	-14.9	1.2	-6.3	10.5	2.2	9.1	2.4	2.8	70	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	36.2	16.8	35.0	16.6	33.7	16.2	18.5	31.8	17.8	30.9	17.3	30.0	4.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.8	23.2	13.1	11.1	23.4	12.2	10.4	23.2	57.9	31.9	55.9	30.9	54.0	30.2	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.4	9.3	8.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-13.2	38.5	3.2	1.4	-15.5	39.5	-17.4	40.3	-19.2	41.1	-21.5	42.1		
			-14.0	20.4	2.9	1.2	-16.1	21.2	-17.8	21.9	-19.4	22.5	-21.6	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	-0.4	2.0	6.9	12.2	17.7	23.5	27.0	27.2	22.1	15.1	7.0	1.3	(6)	
(7)		DBStd	10.31	3.94	3.98	3.97	3.65	3.10	2.90	2.68	2.56	3.21	3.45	3.69	4.41	(7)	
(8)		HDD10.0	1049	322	224	111	18	0	0	0	0	4	101	269		(8)	
(9)		HDD18.3	2610	581	457	354	185	49	2	0	7	107	340	527		(9)	
(10)		CDD10.0	2342	0	0	15	85	239	406	528	534	362	164	11	0	(10)	
(11)		CDD18.3	862	0	0	0	2	30	157	269	276	119	8	0	0	(11)	
(12)		CDH23.3	9259	0	0	1	24	263	1569	3019	3147	1139	97	0	0	(12)	
(13)		CDH26.7	4268	0	0	0	3	40	634	1536	1627	421	8	0	0	(13)	
(14)	Wind	WSAvg	3.5	2.6	3.0	3.7	3.7	3.8	4.3	4.9	4.6	3.6	2.9	2.6	2.4	(14)	
(15)	Precipitation	PrecAvg	246	20	15	27	48	41	9	7	4	5	20	28	21	(15)	
(16)		PrecMax	386	95	38	50	105	81	30	28	21	28	85	70	89	(16)	
(17)		PrecMin	154	0	0	0	14	8	0	0	0	0	2	0	1	(17)	
(18)		PrecStd	55	18	9	12	23	16	10	7	5	6	18	18	19	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	14.0	21.8	26.1	29.8	35.0	38.2	38.1	34.8	27.9	19.0	14.1	(19)	
(20)			MCWB	5.0	6.6	9.2	12.4	14.3	16.5	17.4	17.2	16.2	12.9	9.6	6.5	(20)	
(21)		2%	DB	8.2	11.8	18.1	23.1	27.8	33.2	36.8	36.2	32.2	26.0	16.9	11.0	(21)	
(22)			MCWB	3.6	5.3	8.2	11.0	13.6	15.7	17.3	16.6	15.3	12.4	8.2	5.4	(22)	
(23)		5%	DB	6.9	9.9	15.9	21.0	26.1	31.9	35.0	35.1	30.8	24.1	15.0	9.0	(23)	
(24)				MCWB	3.1	4.4	7.3	10.5	13.2	15.4	16.8	16.4	14.7	11.9	7.8	4.5	(24)
(25)				DB	5.2	8.1	13.9	19.0	24.8	30.2	33.2	33.9	29.1	22.1	13.0	7.1	(25)
(26)				MCWB	2.0	3.3	6.4	9.7	12.8	14.8	16.5	16.2	14.2	11.3	7.2	3.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.6	10.4	13.6	16.3	18.1	19.6	19.1	17.6	14.3	11.3	7.6	(27)	
(28)			MCDWB	9.3	12.7	18.2	22.4	26.0	31.3	34.2	32.4	30.3	23.6	15.3	11.6	(28)	
(29)		2%	WB	4.4	6.1	9.0	12.2	15.0	17.1	18.6	18.4	16.6	13.3	9.9	6.1	(29)	
(30)				MCDWB	7.2	10.0	16.5	19.8	24.3	30.0	32.3	31.5	28.7	22.5	14.0	9.9	(30)
(31)			5%	WB	3.3	4.9	7.9	11.5	14.1	16.4	18.1	17.7	15.7	12.5	8.8	5.0	(31)
(32)					MCDWB	6.1	8.3	14.6	18.6	23.4	28.8	31.3	30.7	27.6	21.4	13.1	7.9
(33)		10%		WB	2.3	3.8	6.8	10.6	13.4	15.7	17.5	17.2	15.1	11.9	7.7	3.9	(33)
(34)					MCDWB	4.9	7.3	12.6	17.3	22.4	27.7	30.3	30.0	26.5	20.3	12.6	6.7
(35)	Mean Daily Temperature Range		MDBR	8.1	9.1	10.1	10.5	11.5	12.3	12.1	12.4	12.7	11.5	9.6	8.1	(35)	
(36)		5% DB	MCDBR	9.3	11.0	12.7	12.9	13.1	13.2	12.9	13.1	13.9	13.8	11.8	10.2	(36)	
(37)				MCWBR	5.7	6.2	6.2	5.5	4.7	4.5	4.4	4.2	5.0	5.8	6.4	6.3	(37)
(38)		5% WB	MCDWB	8.4	9.6	11.4	11.1	12.1	12.6	12.4	12.5	13.1	12.1	9.7	9.0	(38)	
(39)			MCWBR	5.4	5.9	5.9	5.2	4.7	4.6	4.4	4.2	4.9	5.4	5.6	5.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.346	0.401	0.471	0.459	0.434	0.470	0.436	0.387	0.390	0.332	0.307	(40)		
(41)		taud	2.332	2.221	2.077	1.917	1.998	2.106	2.019	2.112	2.236	2.171	2.320	2.364	(41)		
(42)		Ebn at Noon	862	870	851	811	828	848	813	833	857	807	826	839	(42)		
(43)		Edh at Noon	92	118	150	187	176	158	171	152	127	122	93	83	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.44	3.38	4.16	5.22	6.49	7.83	7.63	6.98	5.69	3.90	2.62	2.05	(44)		
(45)		RadStd	0.19	0.36	0.36	0.48	0.33	0.39	0.26	0.24	0.29	0.29	0.31	0.27	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page